

Figure 2 Evidence Package: contour-constrained WTA residual decoding

Quality gate: $\text{abs}(\text{partial Spearman } \rho) \geq 0.5$ within the selected HistoSeg structure.

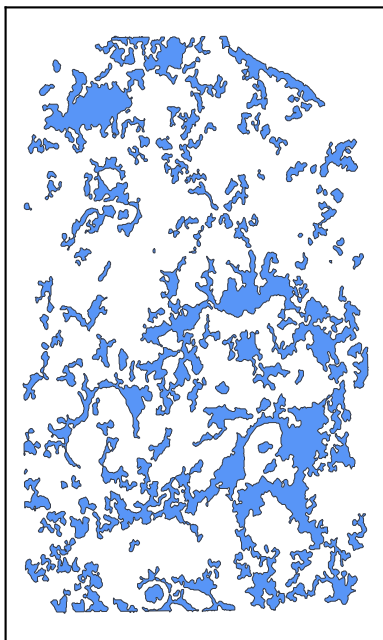
Controls: centroid_x, centroid_y, cell_boundary_distance_um__mean, tile_boundary_distance_px__mean.

- S3 / luminal_estrogen_response / embedding__103__mean: partial $\rho = -0.639$, $n = 157$
- S3 / unfolded_protein_response / embedding__29__mean: partial $\rho = 0.515$, $n = 157$
- S3 / oxidative_phosphorylation / embedding__371__mean: partial $\rho = 0.531$, $n = 157$

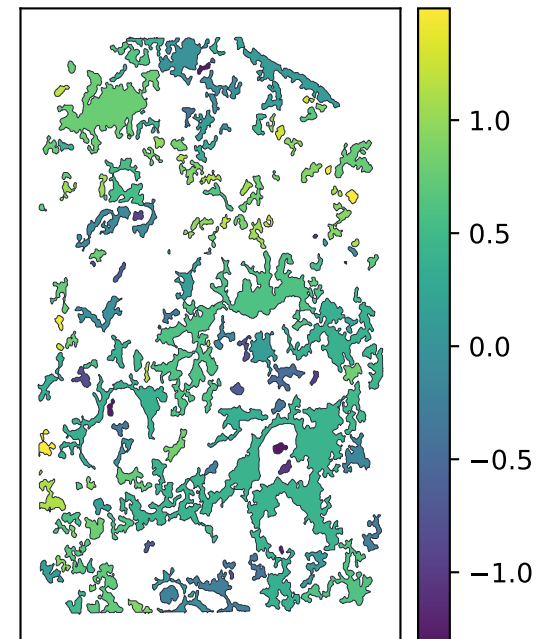
Patch montage source: Figure2_Hero_Patch_Montage.png

S3: luminal_estrogen_response decoded by embedding_103_mean

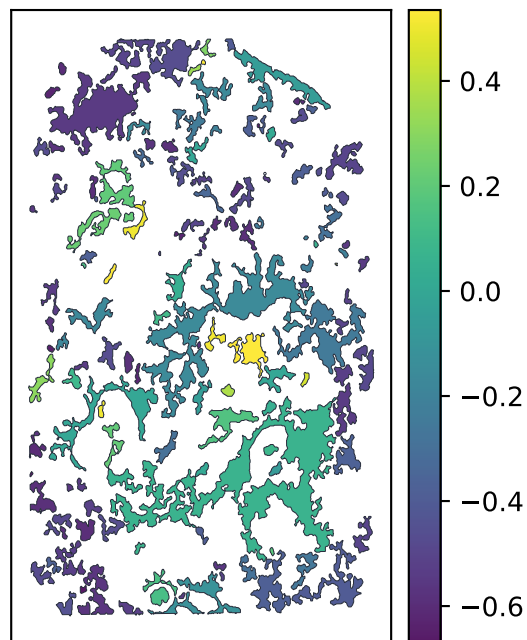
S3 HistoSeg compartment



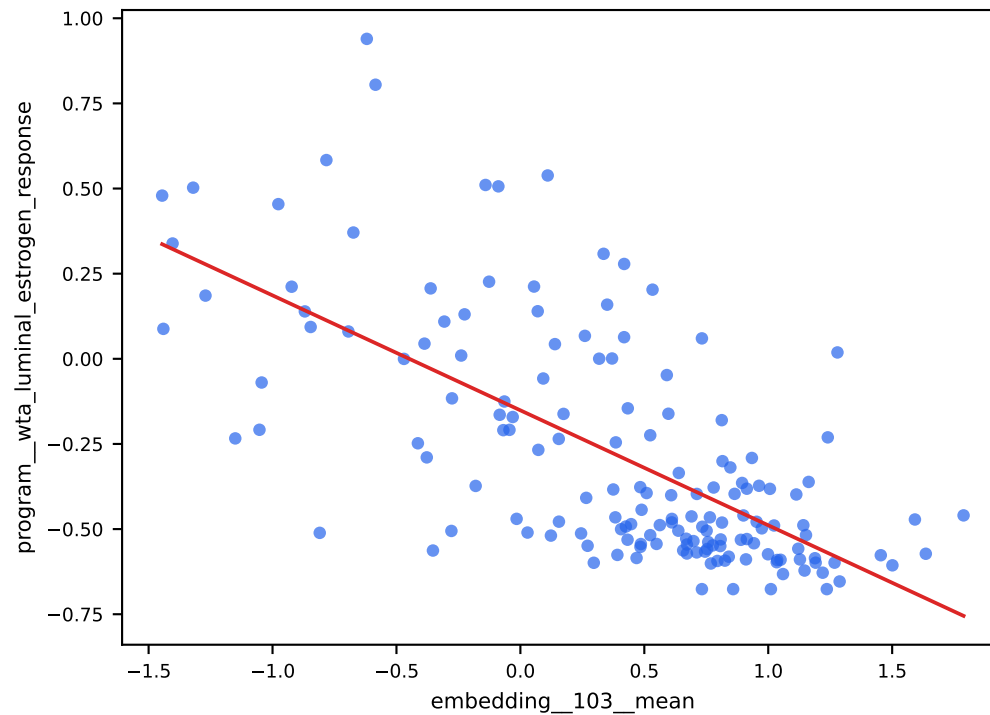
H&E embedding axis



Xenium WTA program

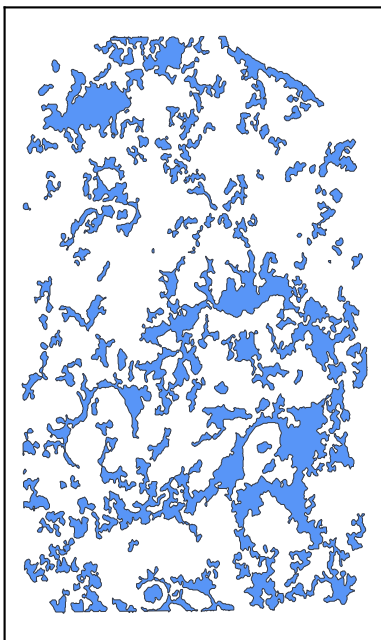


Residual association: $\rho = -0.639$, $p = 4.79 \times 10^{-19}$

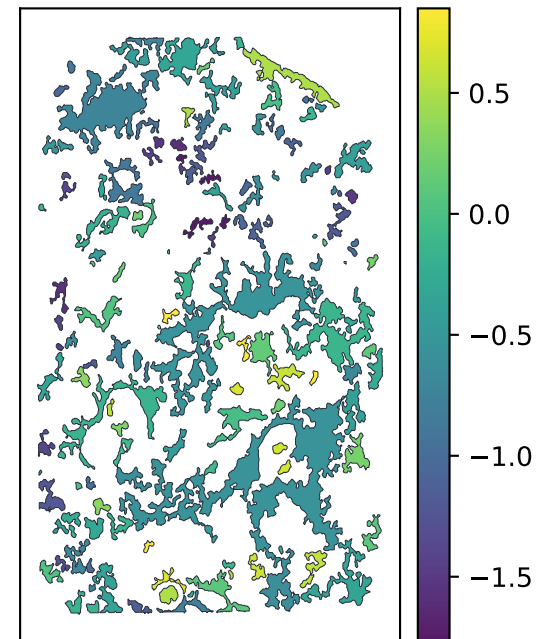


S3: unfolded_protein_response decoded by embedding_29_mean

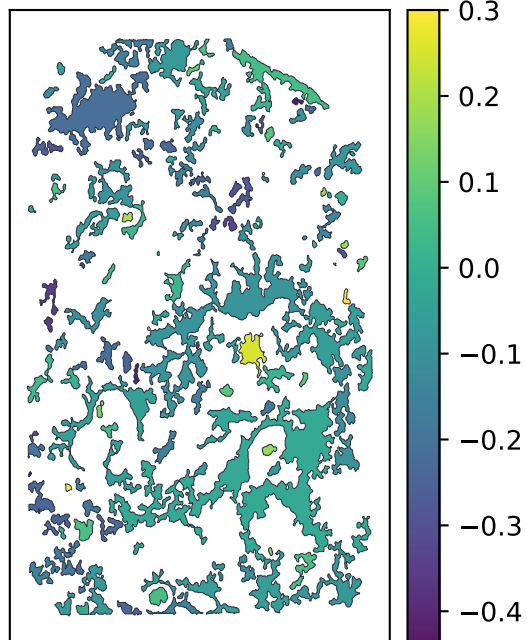
S3 HistoSeg compartment



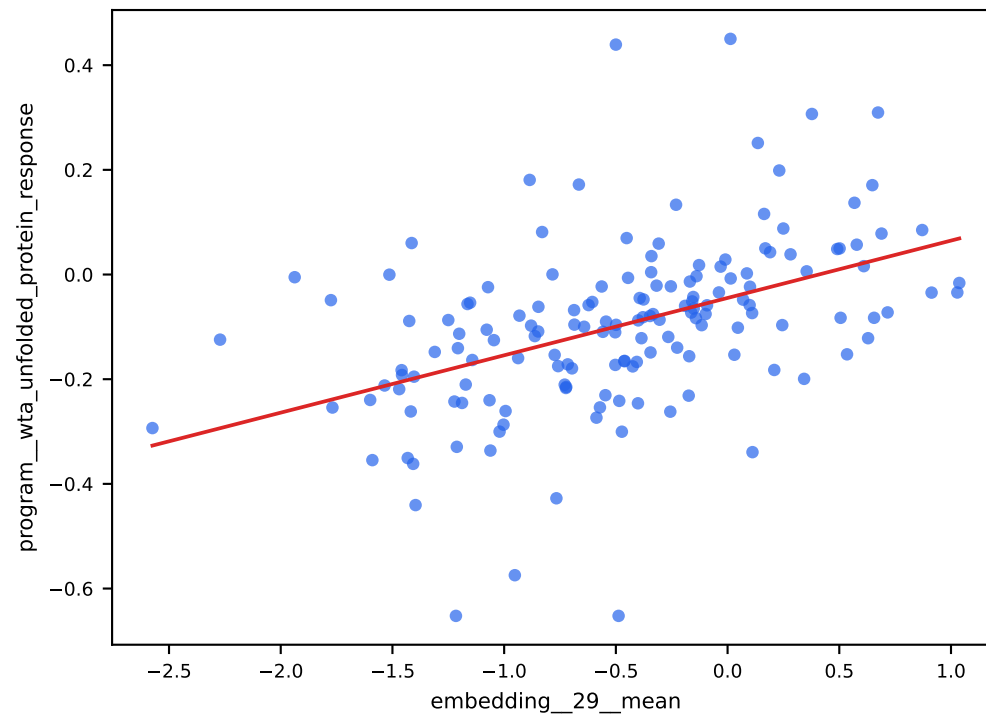
H&E embedding axis



Xenium WTA program

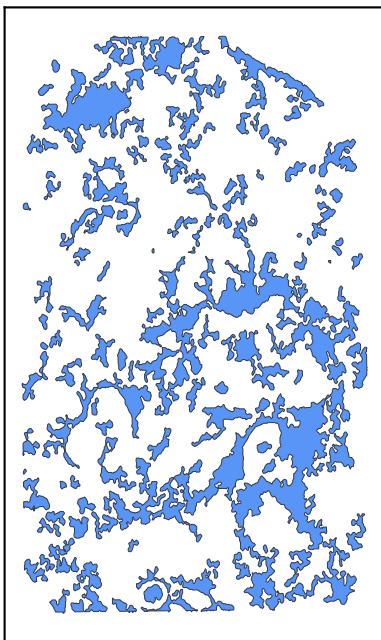


Residual association: $\rho=0.515$, $p=7.91e-12$

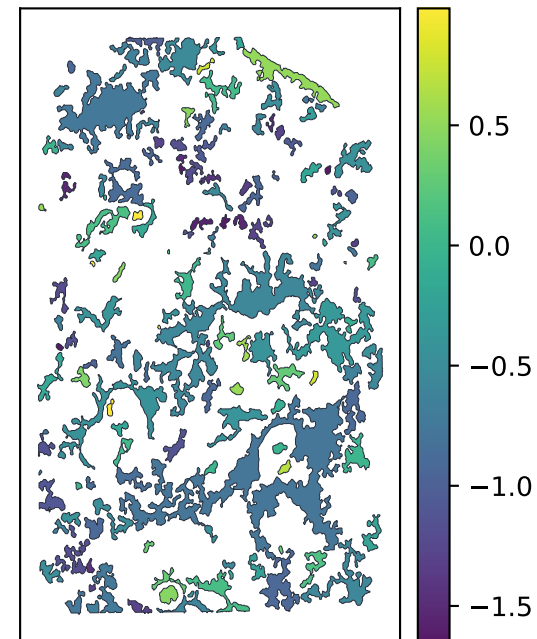


S3: oxidative_phosphorylation decoded by embedding_371_mean

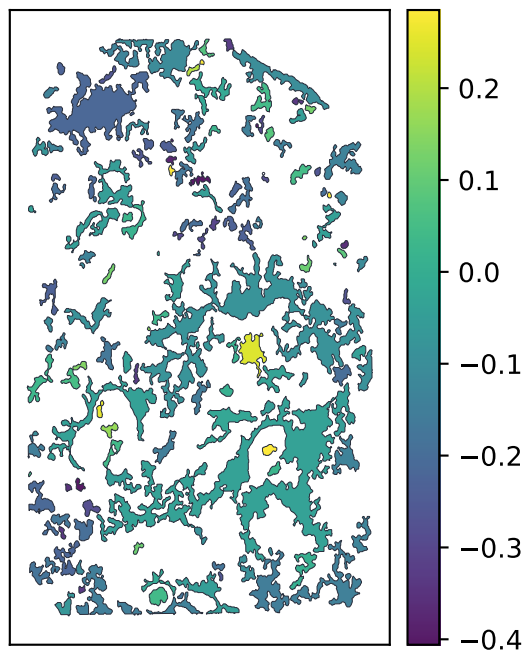
S3 HistoSeg compartment



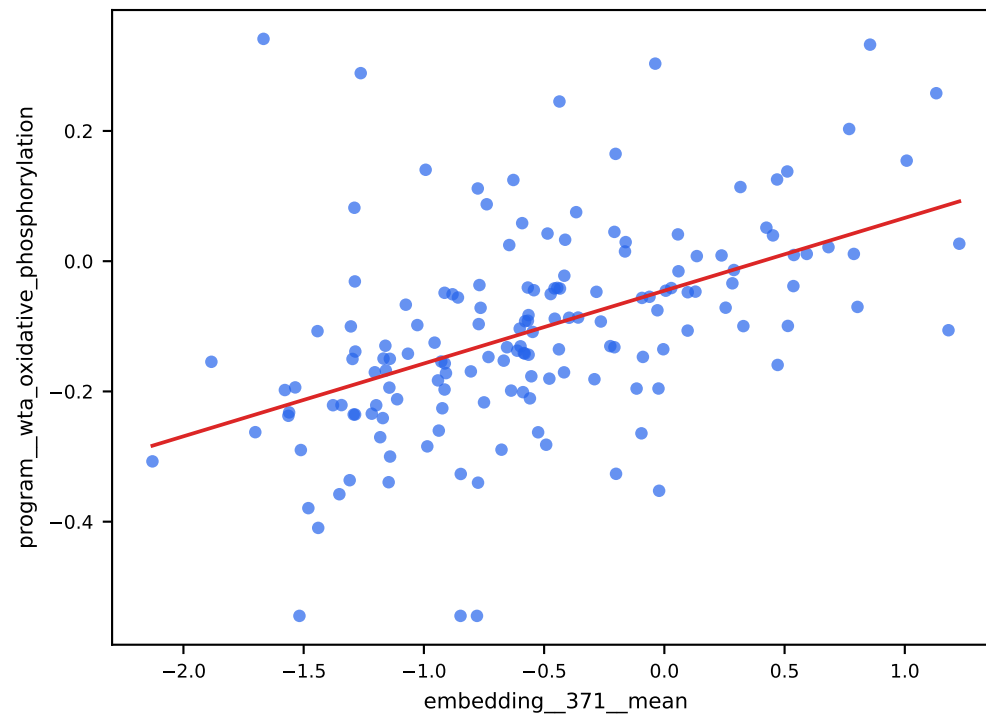
H&E embedding axis



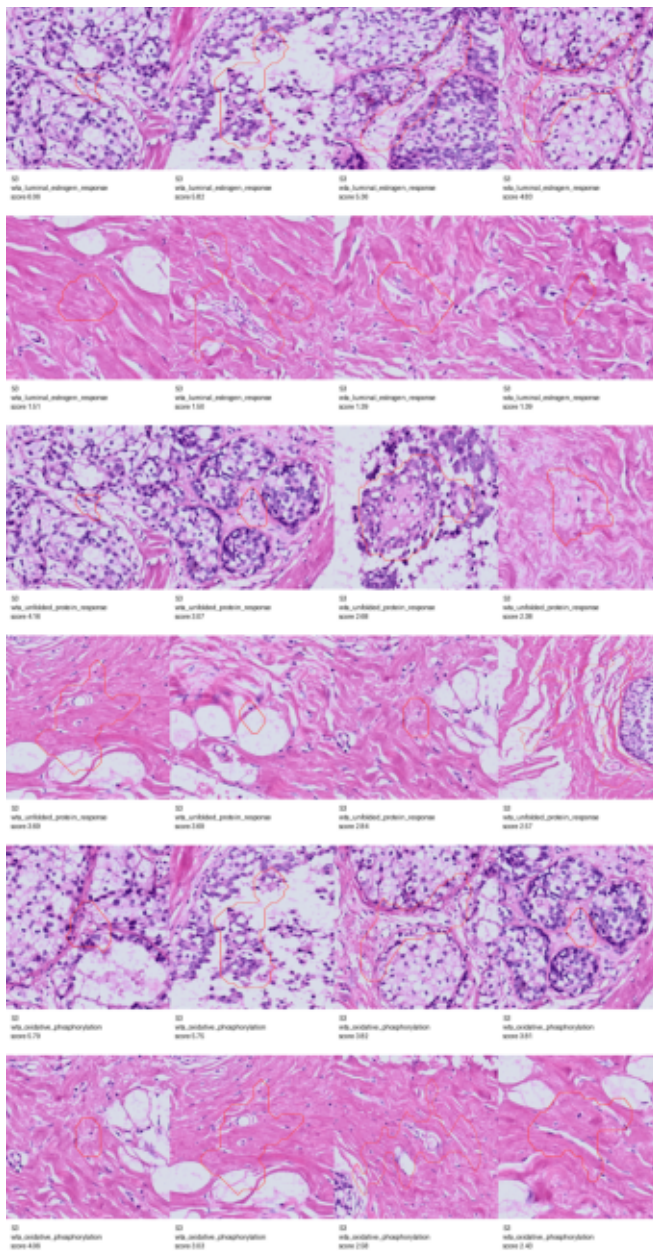
Xenium WTA program



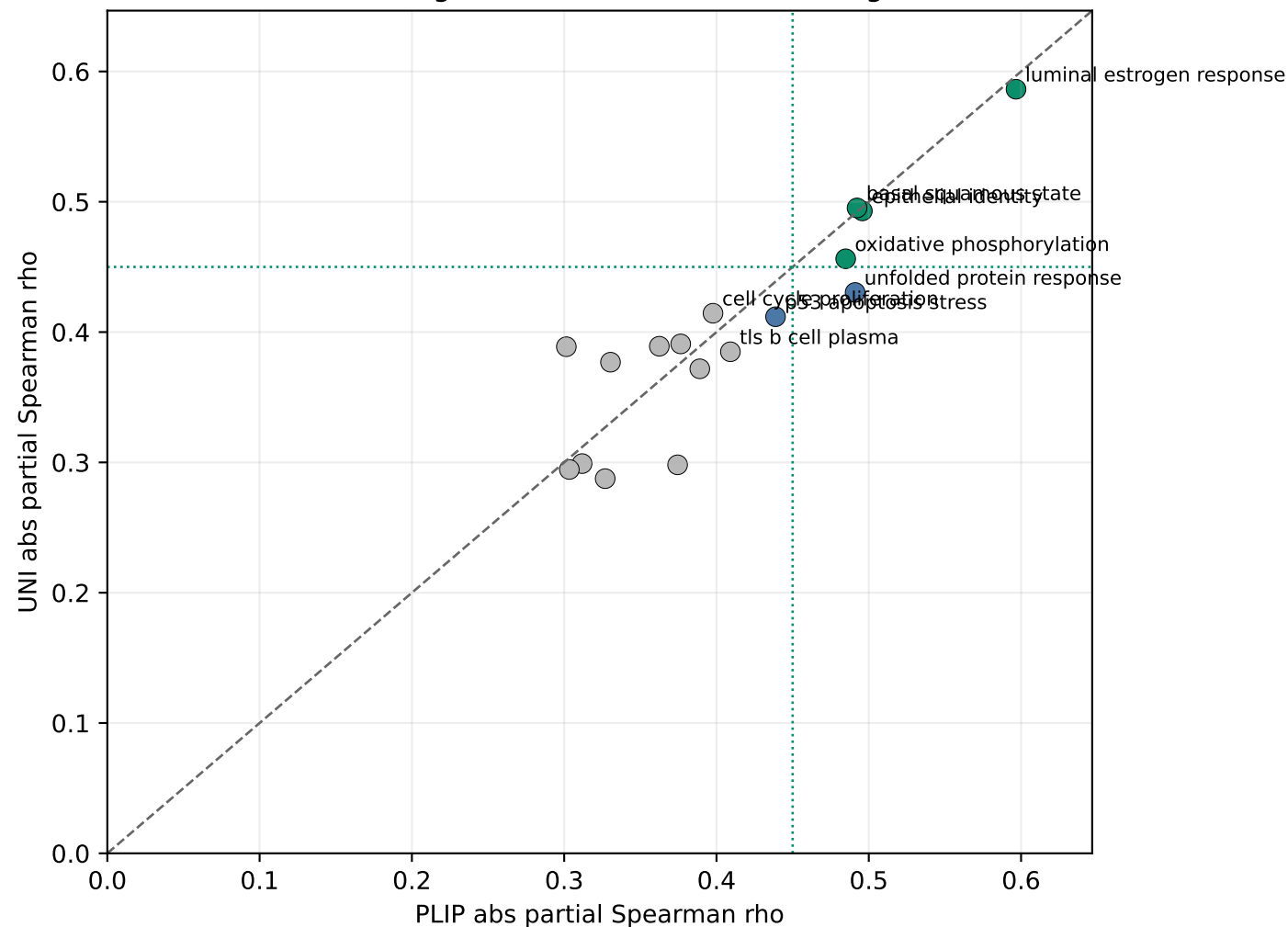
Residual association: $\rho=0.531$, $p=1.38e-12$



H&E hero contours selected by residual morphomolecular concordance



Model-agnostic WTA residual decoding



Top shared WTA programs

